Castles in the Air

Supplement A: More on Dirichlet's proof

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This supplement contains some further information the proof of Dirichlet's theorem which Emily met in Chapter 2, explaining what lies behind some of the imagery in the story. I'll assume some advanced school-level maths here, including some knowledge of Complex numbers, but nothing more.

I should admit at the outset that the ideas here have rather little to do with anything else in the book — this is number theory, not mathematical logic! So if your main interest is in the ideas of mathematical logic, then you may prefer to spend your time exploring the other suplements.

Dirichlet's proof is long and difficult. Some clever alternatives and shortcuts have since been devised by other people, but these still involve many of the same ingredients as the original proof, and there's also something conceptually satisfying about Dirichlet's original approach which is worth seeing. I hope to say enough here to give you a glimpse of how Complex number magic is able to reveal something new and interesting about the prime numbers. I shall be rather selective, majoring on those aspects of the proof that seem to me of particular interest and beauty, and which tie in with points made in Chapter 2 of the book.

As we saw, Dirichlet's theorem concerns the arithmetic progression with starting value a and gap size b, where a and b have no factor in common:

$$a, \quad a+b, \quad a+2b, \quad a+3b, \quad \cdots$$

The question is whether there are infinitely many prime numbers in this progression.

We'll start with something that might look quite far removed from this problem.

Euler's equation

Our starting point is a famous equation discovered by Euler. As we shall see, this equation serves as the fundamental 'bridge' between the world of prime numbers and the world of Complex number theory. The bridge that Dirichlet and Emily crossed actually corresponded to a rather more elaborate form of this equation, which we'll come to below, but we'll start with Euler's version of the bridge, which presents the connection in its simplest form.

Euler's equation looks like this:

$$\left(\frac{1}{1-2^{-s}}\right)\left(\frac{1}{1-3^{-s}}\right)\left(\frac{1}{1-5^{-s}}\right)\cdots \;=\; 1^{-s}+2^{-s}+3^{-s}+4^{-s}+\cdots$$

The first thing to notice here is left hand side consists of infinitely many terms multiplied together, one for each prime number (2,3,5,...). So the left hand side, at least, has some information about prime numbers buried within it. (You can think of this expression as a simpler version of the blue origami construction in the story.) By contrast, the right hand side consists of infinitely many terms *added* together, one for each positive Whole number (1,2,3,4,...). So there's nothing specifically about prime numbers here. (You can think of this as a simpler version of red origami construction.) What's more, the right hand side is in a way a much simpler kind of expression than the left, and it turns out that we can discover a lot about it using Complex number theory.

This fundamental connection between the world of prime numbers and some other world has far-reaching repercussions. Dirichlet's proof made good use of this connection, but the idea was later exploited in a much more profound way by Bernhard Riemann — hence the brief glimpse from the bridge of the so-called *Riemann hypothesis* (which I won't go into here — see Marcus du Sautoy's book *The music of the primes*).

Returning to Euler's equation, you may feel a bit suspicious about the very idea of multiplying or adding *infinitely* many terms. Can it really be done? Is it 'allowed'? Making all of this rigorous does indeed require some technicalities, I won't go into. But broadly speaking, if we take s to stand for a Real number greater than 1, then it turns out (roughly speaking) that the terms on the right hand side will get smaller and smaller, in such a way that we can add infinitely many terms together and still get a finite result. Likewise, it turns out the terms on the left hand side will get closer and closer to 1, in such a way that we can *multiply* all these terms and still get a finite

result. So at least for $s > 1$, both sides of the equation represent well-defined finite quantities, and the equation will say something meaningful.

Let's now try to get an idea of why Euler's equation is true, or at least plausible. We'll need the following result from school maths:

$$\left(\frac{1}{1-x}\right) \;=\; 1 + x + x^2 + x^3 + \cdots$$

This can be applied to each of the terms on the left side of Euler's equation, so that the left side becomes an infinite product of infinite sums!

$$(1 + 2^{-s} + 4^{-s} + \cdots)(1 + 3^{-s} + 9^{-s} + \cdots)(1 + 5^{-s} + 25^{-s} + \cdots)\ \cdots$$

Now imagine what would happen if you tried to 'expand out' this product as you would in school algebra, not worrying too much about all the 'infinities' this involves. You'd get an enormous sum of terms, each of which was got by selecting one term from each of the bracketed expressions. For instance, we might select 2^{-s} from the first bracketed expression, 9^{-s} from the second, 5^{-s} from the third, and 1 from all the others. Multiplying this selection of terms together gives 90^{-s}, corresponding to the fact that the prime factorization of 90 is $2 \times 3^2 \times 5$. The idea is that for *every* possible way of selecting one term from each of the bracketed expressions, we will get a term like this: a term of the form n^s for some Whole number n.[1] What's more, *every* positive Whole number n has a unique prime factorization in this way, so you can see that every possible term n^{-s} will crop up (exactly once) somewhere in this massive expansion. Or to put it another way, our expansion will look exactly like the right-hand side of Euler's equation!

Of course, the argument I've given isn't a rigorous proof of Euler's equation: for that, we'd need some careful justification for why all these infinitary algebraic tricks were 'allowed'. But hopefully I've said enough to let you see the magic that's at work: the connection between something involving prime numbers and something seemingly much simpler.

We can give a glimpse of the general direction we're heading by using Euler's equation to give an alternative proof that there are infinitely many primes (Euclid's theorem). Suppose (for contradiction) that there were only

[1] You might wonder what happens if your selection features infinitely many terms different from 1, e.g. $2^{-s}.3^{-s}.5^{-s}\ldots$. Roughly speaking, we can ignore this possibility, because the result of multiplying such terms together will 'vanish to zero', and so make no contribution to the overall result.

finitely many primes. Then the left side of Euler's equation would be a product of just finitely many terms, and so would have a definite, finite value, even when $s = 1$. On the other hand, the right hand side *doesn't* have a finite value when $s = 1$: we know that the series $1 + 1/2 + 1/3 + 1/4 + \cdots$ diverges. (More rigorously, one can argue that as s tends to 1 from above, the value of the left side tends to a finite value, whereas the value of the right side tends to infinity.) This is a contradiction, so there are infinitely many primes!

This may seem a lot more complicated than Euclid's original proof, but its advantage is that it can be *generalized* in ways that Euclid's can't. Indeed, Dirichlet's proof is basically a massively elaborated version of the above argument, incorporating some additional machinery to filter out all prime numbers that don't belong to our arithmetic progression $a, a + b, a + 2b, \ldots$. The main piece of technology used to perform this filtering is Dirichlet's theory of *characters* — and this is where Complex numbers come in.

Characters

Dirichlet characters were represented in the story by the dances performed by the 'roots of 1'. I'll first say a bit about what they are, before explaining how Dirichlet incorporated them into a modified version of Euler's equation, and how that helps us to solve the original problem.

Mathematically, a character is a special kind of function χ that maps each Whole number n to a Complex number $\chi(n)$. (The symbol χ is the Greek letter *chi*, which is traditionally used for characters.) More specifically, we're interested here in *characters modulo* b, where b is the 'gap size' in our arithmetic progression. We can think of such a character χ as a dance performed by certain Complex numbers to the ticking of the clock: as the clock ticks count through the Whole numbers $0, 1, 2, 3, \ldots$, the corresponding Complex numbers $\chi(0), \chi(1), \chi(2), \chi(3), \ldots$ manifest themselves by making a small jump in the air.

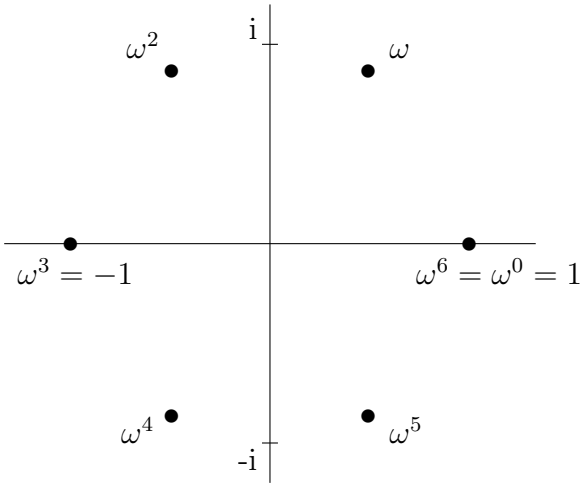
This dance is subject to various rules.

Firstly, the sequence of Complex numbers is one that repeats after every b ste ps, so that $\chi(b) = \chi(0)$, $\chi(b+1) = \chi(1)$, and more generally $\chi(b+n) = \chi(n)$ for every n.

Secondly, it's a rule that $\chi(n) = 0$ if and only if n and b have some factor in common (other than 1). (In particular, if b is a prime number — the easiest case to think about — then $\chi(n) = 0$ if and only if n is a multiple of

b.) In our story, 0 wasn't actually one of the Complex numbers in the dance, so we should imagine that nothing at all happens for these values of n.

Thirdly, it's a rule that if n *isn't* a multiple of b, then $\chi(n)$ is one of the $(b-1)$*th roots of 1* in the Complex plane. These are in fact a set of $(b-1)$ Complex numbers arranged evenly in a circle centred at 0 and including the number 1. For example, if $b = 7$, we get the following six Complex numbers as 'roots of 1': note that they can all be obtained as powers of the one I've labelled ω (i.e. as $\omega^0, \omega^1, \ldots, \omega^5$).



Although these roots of 1 are Complex numbers, Emily came to feel that they weren't so frightening, because there are only finitely many of them, they are spaced comfortably apart in the way that Whole numbers are. In other words, we're working with just a *discrete* set of Complex numbers, and the idea of Complex numbers as continuously varying quantities (which is what Emily found particularly scary) doesn't enter in at this stage.

The final, and most powerful, rule for characters is that they must *respect multiplication*: that is, $\chi(m.n) = \chi(m).\chi(n)$ for all m and n. (The dots here mean multiplication.) This condition puts a very tight constraint on how the dance can proceed, and severely limits the number of such dances that can exist.

Dirichlet proved, in fact, that the number of characters modulo b is exactly the number of numbers in the range $1, \ldots, b-1$ that are *relatively prime to* b (i.e. have no factor in common with b other than 1). This important number is often written as $\phi(b)$. (This part of the proof uses results of Gauss

on modulo arithmetic, represented in the story by the medallions Dirichlet gathered from the rooms close to the Hall of Clocks.)

Although the definition of characters makes sense for any value of b, it will simplify things a bit if we suppose from now on that *b is a prime number*. We'll therefore assume this from now on, although similar ideas work in the general case. In this case, of course, *all* the numbers $1, \ldots, b-1$ are relatively prime to b, so the number of characters modulo b is $b-1$. We'll write these characters as $\chi_0, \chi_1, \ldots, \chi_{b-2}$.

One of these characters (the one we'll call χ_0) is extremely boring: in fact it's defined by setting $\chi_0(n) = 1$ for every n not divisible by p. It's easy to check that this satisfies all the above rules for characters.

Here, for the fun of it, are all six characters in the case $b = 7$, using the names for the roots of 1 given in the picture above. The table shows the values of each $\chi_i(n)$ for $n = 0, \ldots, 6$; after that, each character simply repeats.

	0	1	2	3	4	5	6
χ_0	0	1	1	1	1	1	1
χ_1	0	1	ω^2	ω	ω^4	ω^5	-1
χ_2	0	1	ω^4	ω^2	ω^2	ω^4	1
χ_3	0	1	1	-1	1	-1	-1
χ_4	0	1	ω^2	ω^4	ω^4	ω^2	1
χ_5	0	1	ω^4	ω^5	ω^2	ω	-1

We can do some spot-checks to satisfy ourselves that all these characters respect multiplication: for instance, $\chi_1(3.4) = \chi_1(12) = \chi_1(5) = \omega^5$ (using $12 \equiv 5 \bmod 7$), and $\chi_1(3).\chi_1(4) = \omega.\omega^4 = \omega^5$.

Dirichlet's version of Euler's equation

Here now is Dirichlet's modified form of Euler's equation, with some terms involving a character χ thrown in:

$$\left(\frac{1}{1-\chi(2).2^{-s}}\right)\left(\frac{1}{1-\chi(3).3^{-s}}\right)\left(\frac{1}{1-\chi(5).5^{-s}}\right)\ \cdots$$

$$= \ \chi(1).1^{-s} + \chi(2).2^{-s} + \chi(3).3^{-s} + \chi(4).4^{-s} + \cdots$$

Here χ may be any one of the characters $\chi_0, \chi_1, \ldots, \chi_{b-2}$. Again, the dots mean multiplication.

Before we see what this modification achieves, can we convince ourselves that Euler's equation is still true with all these new $\chi(n)$ terms added? If you try re-running the informal 'proof' of Euler's equation that we gave earlier, you may be able to convince yourself that a similar argument will work provided that $\chi(m.n) = \chi(m).\chi(n)$ for all Whole numbers m and n — which is one of the defining properties of Dirichlet characters as mentioned above. This argument can be rigorously checked, so that the modified version of Euler's equation is indeed true.

As mentioned above, χ can stand for any of the $(b-1)$ different characters, so we actually get $(b-1)$ equations for the price of one. Let me write these equations as

$$P_0 = L_0, \quad P_1 = L_1, \quad \cdots, \quad P_{b-2} = L_{b-2}$$

where P_i is an abbreviation for

$$\left(\frac{1}{1-\chi_i(2).2^{-s}}\right)\left(\frac{1}{1-\chi_i(3).3^{-s}}\right)\left(\frac{1}{1-\chi_i(5).5^{-s}}\right)\ \cdots$$

and L_i is an abbreviation for

$$\chi_i(1).1^{-s} + \chi_i(2).2^{-s} + \chi_i(3).3^{-s} + \chi_i(4).4^{-s} + \cdots$$

Actually, Dirichlet combined all these equations into a single equation, involving *logarithms* of the terms P_i and L_i. This may all look pretty mysterious just now, but at least it's easy to see that this new equation *does* indeed follow from the $(b-1)$ individual equations mentioned above.

$$k_0 \ln P_0 + k_1 \ln P_1 + \cdots + k_{b-2} \ln P_{b-2} \ = \ k_0 \ln L_0 + k_1 \ln L_1 + \cdots + k_{b-2} \ln L_{b-2}$$

Here $k_0, k_1, \ldots$ are certain Complex numbers which I'll say more about later, and ln means 'logarithm to base e'.

I shall call this equation the *Master Equation*: it's really the absolute core of Dirichlet's proof, and it's this equation (rather than the original Euler one) that plays the role of the 'bridge' in our story. The left-hand side of the equation is the big blue origami construction formed by combining the expressions $P_0, P_1, \ldots$, which were themselves each obtained by multiplying infinitely many origami birds together. The right-hand side is the the corresponding red construction that this turns into when we cross the bridge.

What on earth has all of this to do with our original problem? We are about to find out — but in the meantime, it's worth pausing to contemplate just how clever Dirichlet must have been to dream that all of this machinery might be relevant for proving a simple statement about prime numbers!

The structure of Dirichlet's argument

Now for the heart of the proof. For our purposes, it's convenient to break down Dirichlet's argument into three (unequal) parts. We'll start by summarizing each of these in headline form, then go into a little more detail on each of them. Remember that both sides of our Master Equation are actually complicated expressions involving a variable s, which we think of as representing a Real number greater than 1.

First, we show that the right hand side of the Master Equation (the red expression) shoots off to infinity as s gets closer and closer to 1. And because the Master Equation is an equation, that means the same must be true of the left hand side.

Secondly, we study the left side of the Master Equation (the blue expression). We use a bit of algebra and break it down into a series of 'packets', and show that all but one of these packets stays within a finite region as s approaches 1. Which must mean that the one remaining packet shoots off to infinity.

Thirdly — and this is where Dirichlet characters work their real magic — we look closely at what this remaining packet consists of, and show that it is in effect a sum of terms, one for every prime number in our original arithmetic progression $a, a+b, a+2b, \ldots$. Since each of these terms has a finite value at $s = 1$, and we've shown that the value of the whole packet tends to infinity there, it follows that there are infinitely many terms in the packet. So there are infinitely many primes in our original arithmetic progression.

Let's now look at these three steps in a bit more detail.

Part 1: Analysing the red expression

The first step is to show that the value of $k_0 \ln L_0 + k_1 \ln L_1 + \cdots + k_{b-2} \ln L_{b-2}$ tends to infinity as s approaches 1 from above (I'll write this as $s \to 1^+$). (The expression does not, strictly speaking, have a well-defined value when s is exactly 1.) This is what Dirichlet was doing on the far side of the bridge with all those hefty analytic calculations that Emily found so hard to follow — although she was able to grasp the meaning of the above statement with the help of the red billiard ball and its control device.

This is in a way the toughest part of the proof, but I won't say too much about the details here. Dirichlet's original approach went via some complicated but slightly *ad hoc* computations that are perhaps not so conceptually

interesting, and even the later shortcuts that people have found are pretty technical. Nevertheless, the overall outline of the argument is quite easy to present.

Remember that χ_0 and $\chi_{(b-1)/2}$ are special characters, in that their values are always Real numbers (1,-1 or 0). Because of this, it turns out that the corresponding terms L_0 and $L_{(b-1)/2}$ which feature in the above expression will require rather special treatment.

First of all, it's easy to get a good handle on how L_0 behaves as $s \to 1^+$, and in particular, to show that L_0 tends to $+\infty$. This means that the value of $k_0 \ln L_0$ also shoots off to infinity. (Notice that in the Complex plane, a quantity can 'shoot off to infinity' in any direction!) So to show that the value of the *whole* expression shoots off to infinity, all we have to show is that this behaviour of $k_0 \ln L_0$ isn't magically cancelled out by the other terms $k_i \ln L_i$. For this, it's enough to show that each $k_i \ln L_i$ stays within some bounded, finite region of the Complex plane as $s \to 1^+$.

We now need to know just a little bit about the Complex logarithm function. Simplifying a bit, the logarithm $\ln z$ has a well-defined value for every Complex number z except 0; but if we let z tend to 0 from any direction, $\ln z$ will shoot off to infinity. This means that to show that some $k_i \ln L_i$ remains bounded as $s \to 1^+$, all we have to show is that the value of L_i at $s = 1$ is some *non-zero* Complex number z: if this is so, then $\ln z$ will be well-defined, so $k_i \ln L_i$ will simply converge in a nice civilized way to $k_i \ln z$ as $s \to 1^+$, and all will be well.

So the crux is to show that each of $L_1, \ldots, L_{b-2}$ are non-zero when $s = 1$. It may seem that this is a very mild and unimpressive result: after all, there's no reason to expect that these numbers *would* be zero, and it would be an amazing coincidence if it were. But it's a curious feature of mathematics that often it's these very weak results — the ones that intuitively seem 'almost certain' to be true — that are the very hardest to establish by a rigorous proof.

The very hardest part of all is to show that $L_{(b-1)/2} \neq 0$ when $s = 1$: this was the glowing medallion that Dirichlet seemed so pleased with. Once this is in our possession, and the behaviour of L_0 has been analysed, it takes only moderate effort to show that all the other L_i are non-zero at $s = 1$.

The $L_{(b-1)/2} \neq 0$ medallion has an interesting story which I didn't attempt to convey within the book. Dirichlet originally established this result by means of some fearsomely complex and not very illuminating calculations. Very soon after this, however, he made an amazing discovery: the very same

quantity $L_{(b-1)/2}$ also crops up in the investigation of an entirely different problem in number theory, "between which and the subject treated here one would first guess there to be no connection," as he put it. More specifically, in another major piece of work at around the same time, Dirichlet discovered a marvellous formula for the number of essentially different *binary quadratic forms* with a given *determinant* (don't worry about the buzzwords here). This so-called *class number formula* involved a quantity that was just a thinly disguised version of $L_{(b-1)/2}$. So by turning the whole thing round, since we know anyway that the number of quadratic forms can't be zero, we can deduce that $L_{(b-1)/2} \neq 0$: exactly what we need for the hardest part of our proof.

In the story, I portrayed Dirichlet's proof as involving two quite different regions of the Castle. But to take account of this application of the class number formula, one would really have to take a long detour into a third region distinct from either of these: the theory of quadratic forms. Perhaps the number theorists have some deep understanding of 'why' this same quantity $L_{(b-1)/2}$ should crop up in these two quite different contexts; but I confess that for me, this 'coincidence' appears mysterious almost to the point of miracle: an astonishing manifestation of 'patterns woven deeply into the fabric of mathematics as a whole', as Dirichlet says in our story.

Part 2: Manipulating the blue expression

As we noted earlier, because the Master Equation is an equation, we now know that the value of its *left* hand side — namely $k_0 \ln P_0 + k_1 \ln P_1 + \cdots + k_{b-2} \ln P_{b-2}$ — goes to infinity as $s \to 1^+$. Let's now analyse this expression to see what this means. We'll start by considering each of the terms $k_i \ln P_i$ separately.

Remember that P_i is an abbreviation for a certain infinite product of expressions. This is the point at which it starts to emerge why *logarithms* were introduced into the game — because the logarithm of a product is the sum of the logarithms. So we have

$$\ln P_i \;=\; \ln\left(\frac{1}{1-\chi_i(2).2^{-s}}\right) + \ln\left(\frac{1}{1-\chi_i(3).3^{-s}}\right) + \ln\left(\frac{1}{1-\chi_i(5).5^{-s}}\right) + \cdots$$

Now let's look at each term in *this* expression separately. Another result

from (advanced) school maths is that

$$\ln\left(\frac{1}{1-x}\right) \;=\; x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \cdots$$

So each term in the previous equation can be expanded as follows:

$$\ln\left(\frac{1}{1-\chi_i(p).p^{-s}}\right) = \chi_i(p).p^{-s} + \frac{\chi_i(p)^2.p^{-2s}}{2} + \frac{\chi_i(p)^3.p^{-3s}}{3} + \frac{\chi_i(p)^4.p^{-4s}}{4} + \cdots$$

Looking at the terms in this series, we'll the first one (involving p^{-s}) a *first order* term, the second one (involving p^{-2s}) a *second order* term, and so on.

Putting all this together, we can now see that the whole left hand side of the Main Equation can be expanded to a massive 'sum of sums of sums', which I won't write out in detail. It's now a simple matter to break this sum into 'packets' by regrouping it so that all the first order terms are placed in one packet, all the second order terms in another, and so on. So schematically, the left hand side of the Main Equation will get rewritten like this:

$$\text{(big sum of first order terms)} \;+\; \text{(big sum of second order terms)} \;+\; \cdots$$

The next step is to show that the values of the second, third and all higher packets, even when combined, remain within some bounded finite region as $s \to 1^+$. This part of the proof isn't too hard, though I won't give details here. The main idea — taking second order terms as an example — is that the terms p^{-2s} get smaller very quickly as the prime p increases, so that adding all these terms together produces only a finite result. And higher order terms, e.g. those involving p^{-3s} will get smaller more quickly still.

Because the second, third and all higher packets remain finite (even when combined), we can conclude that the 'big sum of first order terms' must shoot off to infinity as s approaches 1. For the final stage of this proof, we shall need to 'open' this packet and see what it consists of.

Part 3: The magic of Dirichlet characters

By working through the algebra for the argument I've just sketched, we can see that the sum of first order terms will look like this:

$$\begin{aligned}
& k_0\left(\chi_0(2).2^{-s} + \chi_0(3).3^{-s} + \chi_0(5).5^{-s} + \cdots\right) \\
+\;& k_1\left(\chi_1(2).2^{-s} + \chi_1(3).3^{-s} + \chi_1(5).5^{-s} + \cdots\right) \\
+\;& \cdots \\
+\;& k_{b-2}\left(\chi_{b-2}(2).2^{-s} + \chi_{b-2}(3).3^{-s} + \chi_{b-2}(5).5^{-s} + \cdots\right)
\end{aligned}$$

Next, let's regroup this so that the terms corresponding to each prime number p — the ones involving p^{-s} — are grouped together. For each prime p, we get an expression like this:

$$p^{-s}.(k_0.\chi_0(p) + k_1.\chi_1(p) + \cdots + k_{b-1}.\chi_{b-1}(p))$$

Now here is where the clock's music works its remarkable magic. Remember that the characters $\chi_0, \chi_1, \ldots$ all represent different dances, so at a typical 'time' t, they are likely to feature many different roots of 1. However, a certain amount of freedom is given to us by the Complex coefficients $k_0, k_1, \ldots$, which we haven't said much about so far. Specifically, we can choose these numbers to 'adjust' the dances so that they all feature the *same* root of 1 at time a, where a is the starting value of our original arithmetic progression we're interested in. More specifically, we *choose* the numbers k_i so that $k_i.\chi_i(a) = 1$ for each i, simply by taking $k_i = \chi_i(a)^{-1}$. Because all the clocks repeat their pattern after b steps, this means that if a prime number p happens to be of the form $a + nb$, we will have:

$$k_0.\chi_0(p) + k_1.\chi_1(p) + \cdots + k_{b-1}.\chi_{b-1}(p) \;=\; 1 + 1 + \cdots + 1 \;=\; b - 1$$

So in this case, the contributions from the various clocks will add up to $b-1$. This is what was represented by strong chords struck by the music on each of the numbers in our original progression.

However, the magic is that if the dances $\chi_0, \chi_1, \ldots$ were forced to agree at times $a + nb$ in this way, it's guaranteed that they would *cancel one another out* at all other times. That is, for any number t *not* of the form $a + nb$ or nb, the values of $k_0.\chi_0(t), k_1.\chi_1(t), \ldots$ would either all be zero (when t is divisible by p) or would be different roots of 1 which would add up to zero, thanks to a certain 'symmetry' of the situation. So if p is any prime *not* in our original progression, the expression written above will vanish entirely. (If you know some physics, you can think of this as a bit like a system of waves that interfere constructively at that values $a + nb$, and destructively everywhere else.)

We're almost done. Our big sum of first order terms has now reduced to a sum of terms of the form

$$p^{-s}.(b - 1)$$

where the values of p are *just* the prime numbers in our original arithmetic progression. And we've established above that this sum must go to infinity

as $s \to 1^+$. On the other hand, it's easy to see that each individual term $p^{-s}.(b-1)$ approaches the perfectly finite value $p^{-1}.(b-1)$ as $s \to 1^+$. So we're forced to conclude that there are infinitely many primes in our arithmetic progression. Q.E.D.!

This is a good example of a clever proof, because it's very hard to see what might have led Dirichlet to conceive all these extraordinary detours in order to prove a relatively simple fact about Whole numbers. But somehow or other, they work — and that's the beauty and mystery of mathematics for you.